

- Process the impulse signal using your bandPass.m function
 - Plot the result using **freqz(output)**
- Process the impulse signal using your bandStop.m function
 - Plot the result using **freqz(output)**

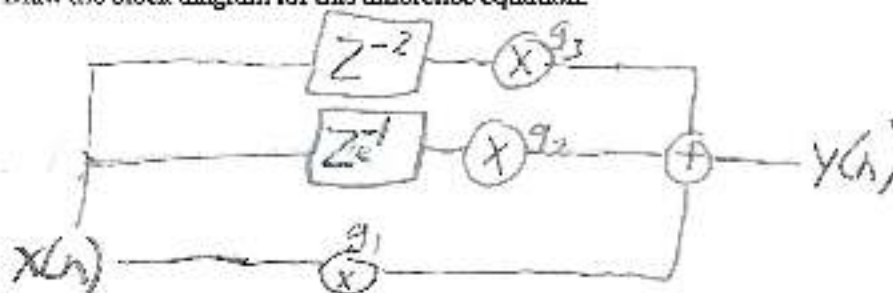
Remember to add comments to your code to explain what each command is accomplishing.

3 SPECTRAL ANALYSIS OF SYSTEMS

Given the following system, analyze its spectral characteristics with the frequency response and pole-zero plot. Refer to the following difference equation:

$$y[n] = (0.45) * x[n] + (-0.5) * x[n-1] + (0.05) * x[n-2]$$

Draw the block diagram for this difference equation:



Take the Z - transform:

$$\bar{Y}(z) = .45\bar{X}(z) + -.5\bar{X}(z)z^{-1} + .05\bar{X}(z)z^{-2}$$

Re-write the result in the form of a transfer function $\frac{Y(z)}{X(z)} = H(z)$:

$$\bar{Y}(z) = \underbrace{.45\bar{X}(z)}_{\text{Factor } \bar{X}(z)} + \underbrace{-.5\bar{X}(z)}_{\text{Factor } \bar{X}(z)}z^{-1} + \underbrace{.05\bar{X}(z)}_{\text{Factor } \bar{X}(z)}z^{-2}$$

$$\frac{\bar{Y}(z)}{\bar{X}(z)} = \frac{\bar{X}(z)}{\bar{X}(z)} (.45 - .5z^{-1} + .05z^{-2})$$

$$\rightarrow H(z) = .45 - .5z^{-1} + .05z^{-2}$$